CivicPulse - Data Quality Report

1. Executive summary

The dataset is fit for analysis: 14 of 15 columns are at least 96.8% complete, all 45,750 records were loaded without loss, only 10 records (0.02%) have impossible dates, and 331 (0.72%) are marked Closed without a closing date, all of them from a single agency, the Department of Homeless Services (DHS).

- **One major limitation:** `due_date` is 99.5% empty, so deadline compliance cannot be measured. It is replaced by response-time targets (share of requests closed within 24 hours and within 7 days).
- **One sampling bias detected and corrected** before analysis: the first extraction over-represented night-time requests (64% between 0:00 and 2:00).
- **All other issues are minor** (under 3.5% of records each) and handled by explicit rules, listed in section 5.

2. Scope

The assessment covers the full sample used by CivicPulse, both as extracted and as loaded into PostgreSQL.

Item	Value
Source	NYC Open Data — *311 Service Requests from 2020 to Present* (`erm2-nwe9`), via the Socrata API
Period	1 January – 31 December 2024 (366 days)
Extraction date	6 November 2026
Sample	45,750 requests: about 2% of 2024 requests (`unique_key` ending in 07 or 57), then a random draw with a fixed seed
Columns assessed	15: `unique_key`, `created_date`, `closed_date`, `due_date`, `agency`, `agency_name`, `complaint_type`, `descriptor`, `status`, `resolution_description`, `incident_zip`, `borough`, `latitude`, `longitude`, `open_data_channel_type`
Database	PostgreSQL, database `nyc311`, 4 tables (`service_request`, `agency`, `complaint_type`, `location`)

3. Method

The data was assessed on four standard quality dimensions, at two stages: on the extracted file (`02_check_sample.py`) and in the database (`data_quality_checks.sql`).

Dimension	Question	Checks performed
Completeness	Are the values present?	Share of empty values per column

Dimension	Question	Checks performed
Validity	Are the values possible?	Closing date before creation date; borough outside the 5 boroughs
Integrity	Did the data survive loading intact?	Rows in file vs rows in database; every request linked to an agency, complaint type and location (foreign keys)
Representativeness	Does the sample reflect real activity?	Distribution by month, hour of day, borough and complaint type

4. Results

All four dimensions pass, with one exception on completeness (`due_date`).

4.1 Completeness

Eight columns are 100% complete: `unique_key`, `created_date`, `agency`, `agency_name`, `complaint_type`, `status`, `borough`, `open_data_channel_type`. The other seven:

Column	Missing	Assessment
`due_date`	99.5%	Unusable
`descriptor`	3.2%	Acceptable
`closed_date`	1.7%	Acceptable
`resolution_description`	1.5%	Acceptable (not analyzed)
`latitude`	1.5%	Acceptable
`longitude`	1.5%	Acceptable
`incident_zip`	0.9%	Acceptable

4.2 Validity

Rule	Records breaking it	Share
A request marked `Closed` must have a `closed_date`	331 (all DHS)	0.72%
`borough` must be one of the 5 boroughs	28 (`Unspecified`)	0.06%
`closed_date` must not be before `created_date`	10 (9 DOT Pending, 1 NYPD Closed)	0.02%

4.3 Integrity

All 45,750 records in the file were loaded into `service_request` (0 lost), and every record is linked to an existing agency, complaint type and location through its foreign keys.

4.4 Representativeness

The corrected sample follows a realistic pattern on every axis checked.

Axis	Observation
Month	All 12 months present, 3,150 (February) to 4,179 (December) requests
Hour of day	Peak 9:00–11:00 (about 2,700 requests per hour), low 3:00–5:00 (about 500)
Borough	Brooklyn 13,738 · Queens 10,954 · Bronx 9,873 · Manhattan 9,575 · Staten Island 1,582
Complaint type	Illegal Parking first (6,700), then Noise - Residential (4,977) and Heat/Hot Water (3,480)

5. Issue log

Each issue has a fixed treatment rule, applied the same way in every query and in the Power BI views.

#	Issue	Severity	Affected	Treatment
1	First sample over-represented night-time requests	High (fixed)	Whole first sample	Re-extracted with a key-based random sample; verified by hour distribution
2	`due_date` empty	High	45,499 (99.45%)	Column not used; "overdue" KPI replaced by % closed within 24 h and within 7 days
3	Requests that took more than 90 days to close	Information	1,792 (3.92%)	Kept: they are real; the median is used so they do not distort the typical time to solve
4	`descriptor` empty	Low	1,483 (3.24%)	Stored as `Unspecified`; request still counted under its complaint type
5	`closed_date` empty	Low	792 (1.73%)	Excluded from response-time KPIs; counted as open in backlog KPIs when status is not `Closed`
6	`latitude` / `longitude` empty	Low	691 (1.51%)	Excluded from maps only
7	`incident_zip` empty	Low	415 (0.91%)	Stored as `Unknown`; still counted in its borough
8	Status `Closed` but no `closed_date` (part of issue 5)	Low	331 (0.72%), all from DHS: 331 of its 685	Excluded from response-time KPIs; still counted as closed, since the status says so. DHS solving time and % within 24 h / 7

#	Issue	Severity	Affected	Treatment
			requests (48%)	days cover only its 354 requests with a closing date, so they are shown as partial
9	`borough = Unspecified`	Low	28 (0.06%)	Excluded from borough comparisons
10	`closed_date` before `created_date`	Low	10 (0.02%): 9 DOT requests with status Pending, 1 NYPD request with status Closed	Excluded from response-time KPIs; the 9 Pending requests still count as open in backlog KPIs

No record was deleted from the database: issues are handled in the queries, so the raw facts stay available for audit.

6. Impact on the analysis

Every planned analysis can be delivered except deadline compliance, which is replaced by response-time targets.

Analysis	Status	Note
Demand by month, hour, weekday	Supported	Dates 100% complete
Demand by borough and complaint type	Supported	28 `Unspecified` records excluded from borough views
Response time by agency and complaint type	Supported	Median on about 98% of requests with a valid closing date; DHS is partial (354 of its 685 requests)
Open backlog	Supported	Based on `status`, 100% complete
Maps	Supported	98.5% of requests geolocated
Deadline compliance (overdue requests)	Not supported	Replaced by % closed within 24 h and 7 days

Interpretation limit: the sample is about 2% of 2024 requests, so results are reported as shares, rankings and medians, not as the city's total volumes.